

LLM Evaluation Report

1. Inter-Annotator Agreement (Data Quality)

Based on 96 overlapping articles.

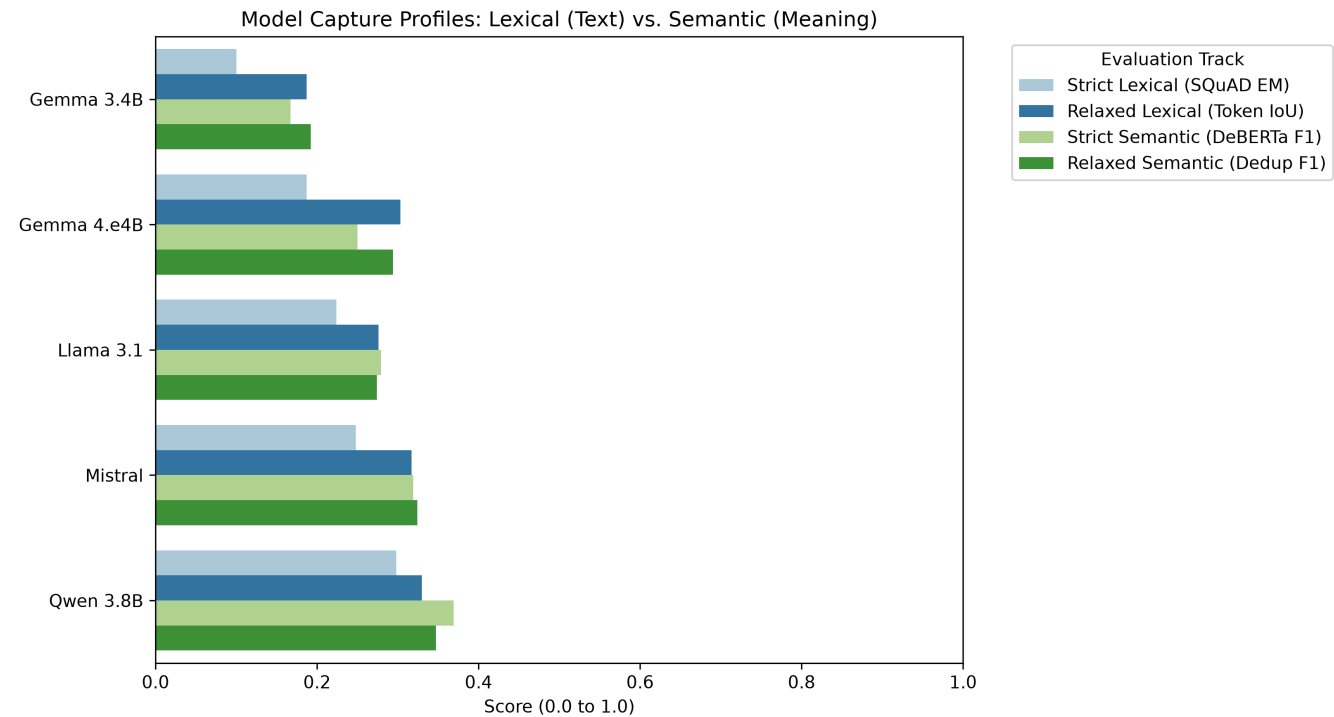
Metric	Score
Krippendorff's Alpha (Ordinal)	0.9051
ICC(2,k) (Absolute Agreement)	0.9665

2. Metric-to-Human Correlation

Total Framework Samples: 96

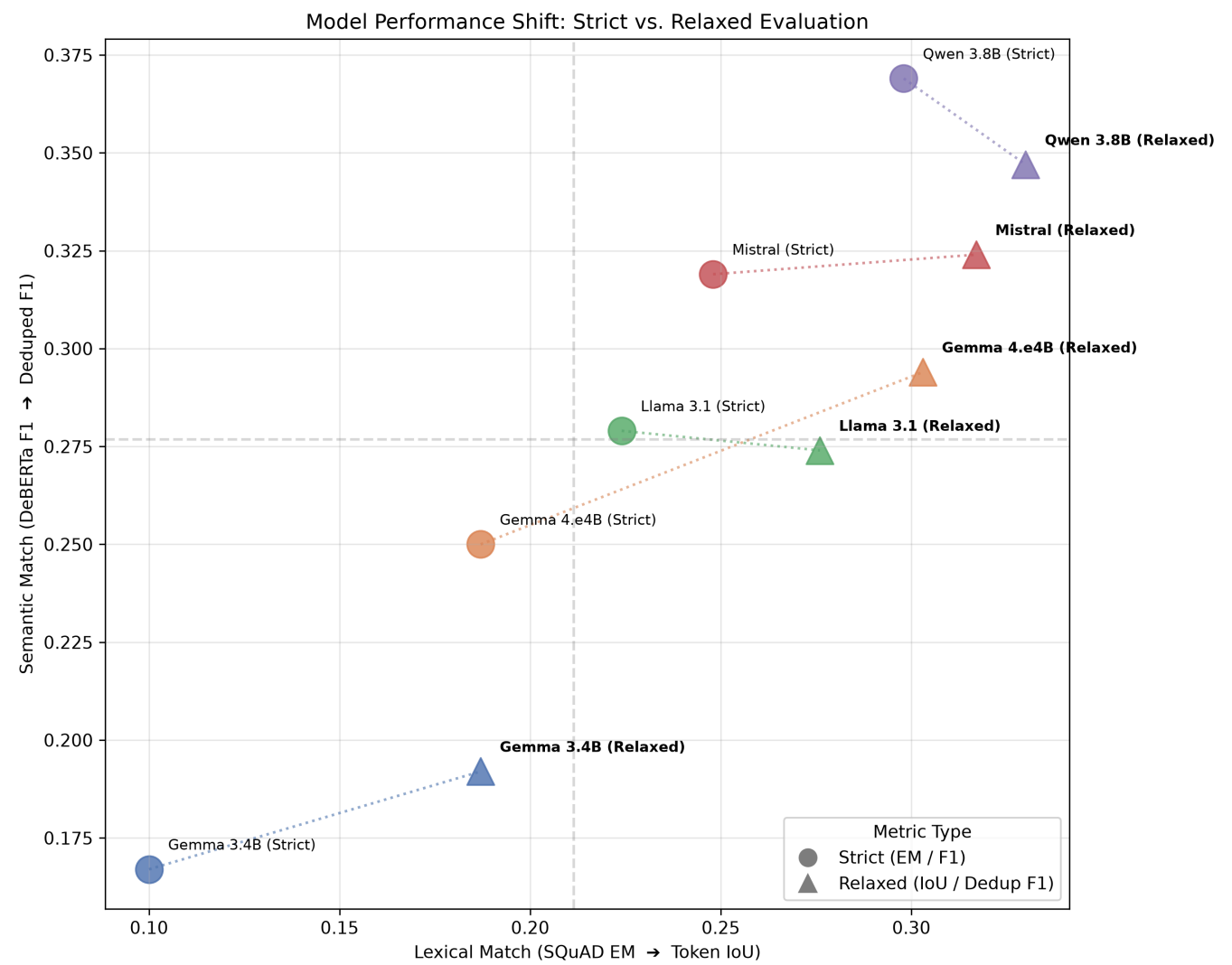
Correlation Comparison: Strict vs. Relaxed Metrics

Evaluation Dimension	Strict Metric	Relaxed Metric	Strict Spearman (ρ)	Relaxed Spearman (ρ)
Lexical Match	SQuAD EM	Token IoU	0.7950	0.8242
Semantic Match	DeBERTa F1	Deduped DeBERTa F1	0.8178	0.8265



Side-by-Side Ranking Comparison

Rank	Human Preference	Strict Machine Preference (F1)	Relaxed Machine Preference (Dedup F1)
#1	Qwen 3.8B (0.346)	Qwen 3.8B (0.369)	Qwen 3.8B (0.347)
#2	Llama 3.1 (0.285)	Mistral (0.319)	Mistral (0.324)
#3	Mistral (0.193)	Llama 3.1 (0.279)	Gemma 4.e4B (0.294)
#4	Gemma 4.e4B (0.183)	Gemma 4.e4B (0.250)	Llama 3.1 (0.274)
#5	Gemma 3.4B (0.118)	Gemma 3.4B (0.167)	Gemma 3.4B (0.192)



4. Error Analysis & Top Disagreements

Evaluated on 45 complex edge-case rows.

Human Baseline Agreement (Complex Subset)

Metric	Score
Krippendorff's Alpha	0.8323

Metric	Score
Intraclass Correlation ICC(2,k)	0.9380

Strict vs. Relaxed Metric Correlation

Evaluation Dimension	Strict Metric	Relaxed Metric	Strict Spearman (ρ)	Relaxed Spearman (ρ)
Lexical Match	SQuAD EM	Token IoU	0.7138	0.6413
Semantic Match	DeBERTa F1	Deduped DeBERTa F1	0.8066	0.6668

Delta Analysis (Top 5 Disagreements after Relaxation)

Disagreement #1 (Score Delta: 0.7276)

- ID: EVAL_088
- Ground Truth: airstrike
- AI Prediction: air strike
- Scores: Human = 1.0000 | Relaxed BERTScore = 0.2724

Disagreement #2 (Score Delta: 0.5500)

- ID: EVAL_005
- Ground Truth: shells | shelling | bombing
- AI Prediction: shelling | shelling | shelling | shelling | shelling | shelling | shelling | shelling | shells
- Scores: Human = 0.2500 | Relaxed BERTScore = 0.8000

Disagreement #3 (Score Delta: 0.4404)

- ID: EVAL_050
- Ground Truth: bombings | shells | shelling | shelling | shelling | shelling | shelling
- AI Prediction: shells | heavy machineguns | heavy machineguns | shelling | shelling | shelling | shelling | shelling | bombings
- Scores: Human = 0.4167 | Relaxed BERTScore = 0.8571

Disagreement #4 (Score Delta: 0.3334)

- ID: EVAL_072
- Ground Truth: aerial bombardment | ieds
- AI Prediction: IEDs | fired | bombardment | bombardment | bombardment | aerial bombardment | Aerial bombardment
- Scores: Human = 0.3333 | Relaxed BERTScore = 0.6667

Disagreement #5 (Score Delta: 0.3334)

- **ID:** EVAL_025
 - **Ground Truth:** air strikes | explosive barrels
 - **AI Prediction:** warplanes | Warplanes | air strikes | explosive barrels | bombardment
 - **Scores:** Human = 0.3333 | Relaxed BERTScore = 0.6667
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